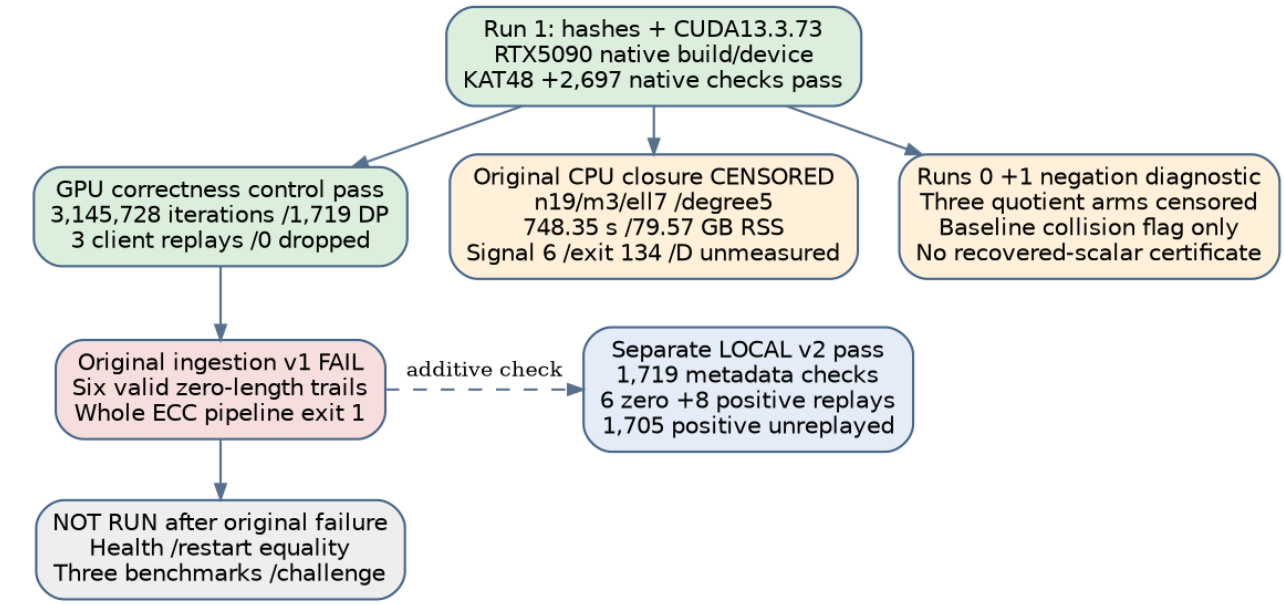
Bounded GPU continuation: actual run 0 and run 1

Audit items 7 and 13; recorded 18 October 2026

GPU control executed; continuation stopped; original closure censored. Run 1 built CUDA13.3.73 and passed native/device checks on RTX5090. Its original ingestion gate then refused valid zero-length trails. Health, restart equality, throughput benchmarks and challenge collection did not run. The original high-memory CPU closure failed allocation without measuring D.



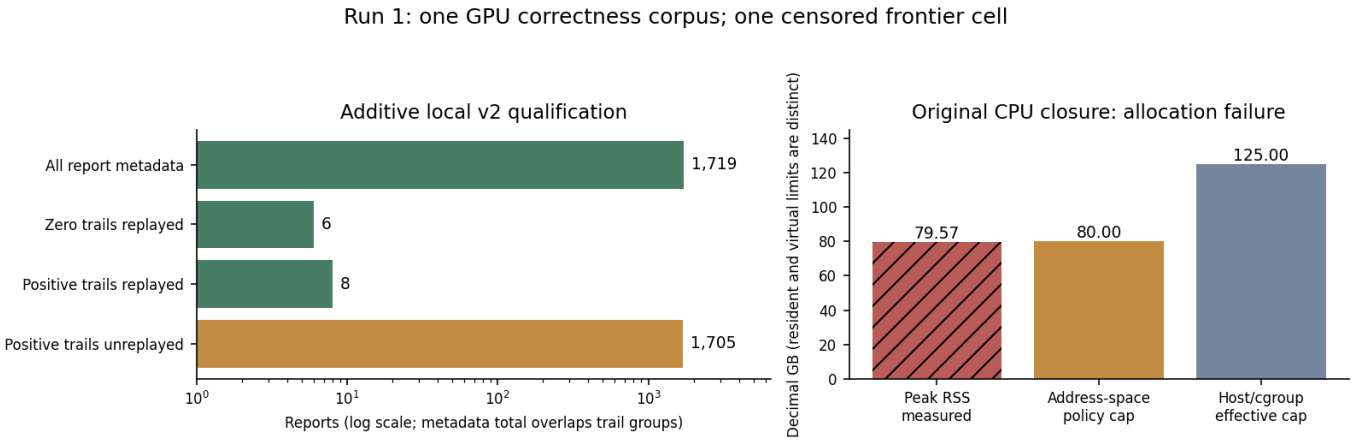
Recovered evidence. Run 0 archive: 15,160 bytes, SHA256 prefix `26d26efe1f6e8d99`. Run 1: 139,739 bytes, SHA256 prefix `c7ddcdbd1a88c72e`. All 147 run1 manifest files passed independent hash verification in both retained and freshly extracted trees. Full hashes, raw commands, resources, failures and source references are in `RUN0_RESULTS.md`, `RUN1_RESULTS.md` and `diagnostics/run1-audit/receipt.json`. The controller records deletion and verified absence of the dedicated pod. No later allocation is represented as a scientific run here.

Identity and runtime. ECC source is pinned to `0ea8931d21d567f83de47c30bdaac0585f479c11` (39 files), public P/Q on $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{131}}$, exact native permuted type-II normal basis and subgroup order 6805647338418769269323201294934099851 29. The polynomial-basis registry alias is not substituted. Delivery source: `9bb0c292b3b1d6b0fafe9c33f73a2145d3dc4830`. Observed driver 580.65.06, driver API 13000, runtime 13030. Native doctor/build passes support this actual configuration, not universal driver compatibility.

GPU correctness and additive local ingestion

Remote host bundle: 2,862 sigma checks including 48/48 public known answers; 28,641 native-CLMUL and28,638 portable controls. Separate native sigma checker: 2,697 checks and 48/48 KAT. The GPU control used run 53180, DP weight 46, 256 threads,16 slots,96 steps and 8 launches. It reported 3,145,728 iterations, 1,719 reports,3 native golden-model replays,0 bad reports and 0 dropped. The raw 0.213-second interval is a short correctness diagnostic, not a repeated campaign benchmark or replacement for historical production rates.

The original independent ingester exited 1 with "invalid trail length." Six retained reports have zero length because their seed start is already DP46. The original remote pipeline remains failed. A separately frozen local v2 qualification preserves v1 and the run's bytes, checks all 1,719 paired reports for public-curve/weight/namespace/encoding, and independently replays *all* 6 zeros plus 8 eligible positive trails. Exactly 1,705 positive trails remain unreplayed. Seed-tampered zero and coordinate-tampered positive controls fail. This local correction does not execute any missing remote gate.



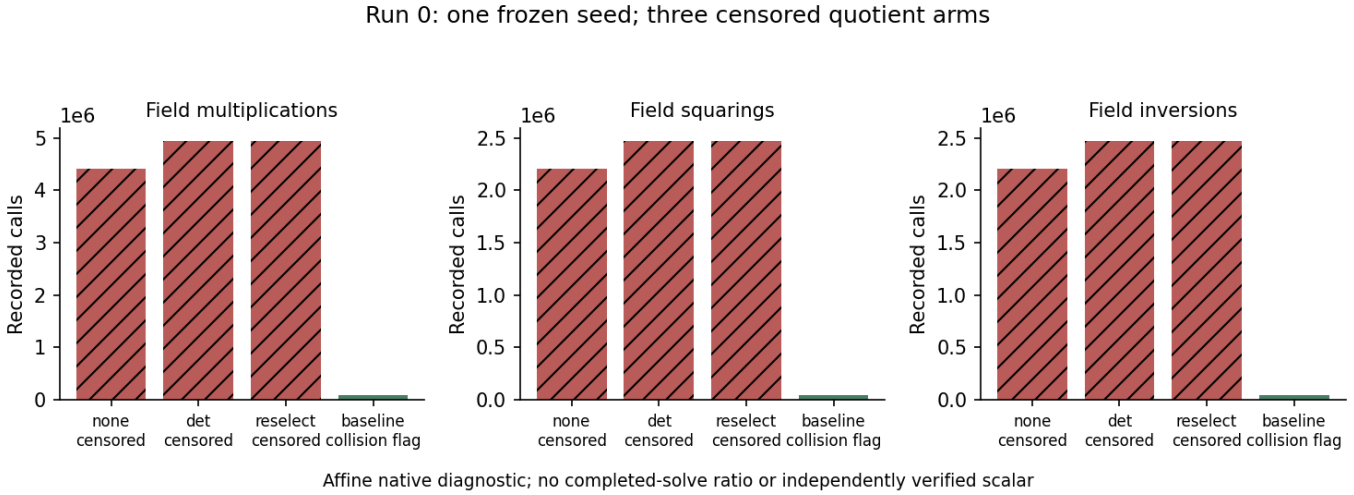
Original remote ingestion failed; local correction does not pass the pipeline. Frontier: 748.35 s, signal 6 / exit 134, D unmeasured.

Original memory-frontier attempt. Committed n19/m3/ell7,draw0, 40 Boolean variables,degree 5,max-columns 800000; native XL/MutantXL closure. Tiny controls passed: SAT retains 12 solutions through 4, UNSAT refutes at 4. Effective host limit 124,999,999,488bytes passed the96GB gate. The process address-space cap was 80,000,000,000bytes. After 748.35 s wall, 700.06 s user and 49.23 s system CPU, allocation of 3320 bytes failed. Peak RSS 77,709,264 KiB is 79,574,286,336 bytes. No frontier JSON result was emitted; D remains **unmeasured**. No bound on D follows from the failed incomplete scan.

The time log simultaneously says "Command terminated by signal 6" and `exit_status:0` from its `%x` field. The separate process exit and terminal receipts say 134. The bare exit-field 0 does not encode signal termination and is not success; signal 6/actual exit 134 governs the failure. Both original fields remain retained. This was CPU closure on a GPU host, not GPU/distributed F4 or an F4 solving-degree measurement; there was no inner closure checkpoint.

Run 0 failure, negation diagnostics and remaining criteria

Run 0 preserved a packaging failure: omitted protocol/binding documents and nonexecutable frontier script. All 39 ECC source hashes passed, then hardware manifest gate exited 1; ECC and frontier were skipped. Four native negation commands executed. The source packet was corrected additively for run1; run0's failures remain charged and immutable.



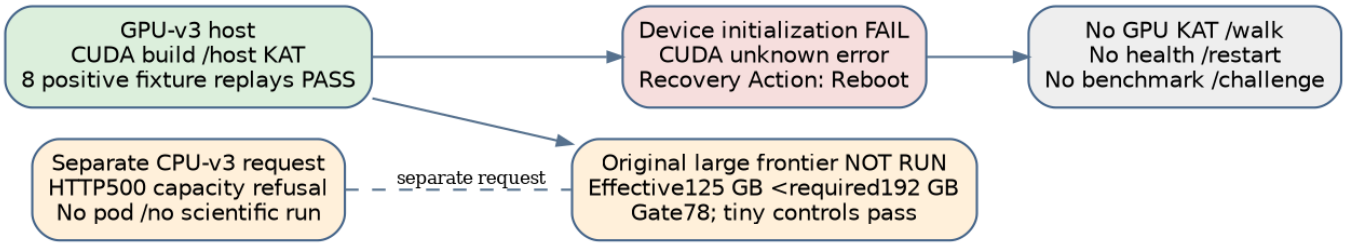
Run 1 repeats the same diagnostic step/operation/DP/censoring/collision counters (only deterministic-break peak RSS differs). All three quotient arms hit the 2,208,379-step budget and remain censored. Baseline reports 42,678 steps and a collision flag without an independently replayed recovered scalar. A successful command exit is not a completed mathematical solve. The original H-RHO-bc6ea6/EXP-RHO-60da4f already has substantial prior trials and retains its affine/Jacobian deviation; this four-arm diagnostic does not close the full experiment. No completed-solve ratio or M-equivalent gain is inferred.

Remaining gates. A prospectively admitted whole-pipeline rerun with the corrected checker must pass health, exact checkpoint-restart equality, three bounded benchmarks, challenge slices and independent ingestion. Run 53181 remains quarantined, without a global seed lease or shared-store upload. Long trails require explicit replay scope; collision detection is not scalar recovery. Finishing ECC2K-130 requires an independently verified challenge scalar. The original frontier still needs completion with a valid resource boundary; a smaller substitute cannot discharge that obligation. Distributed/out-of-core/GPU F4 and the complete negation contract remain open.

Graph and claim boundary. Updated graph sources/renderings are `reports/run-flow.dot` and `reports/plot_run1.py`; native-derived counts are retained in `reports/run1-summary.json`. Canonical sources `papers/ecdlp-constant/sec-ecc2k-campaign.tex` and `sec-ecc2k-gpu.tex` were checked. Their historical production rates, IC leaderboard and forecasts remain current: a short DP46 control and failed closure are not replacement production measurements. No attack-cost, asymptotic, speedup, full challenge or scientific state transition is claimed.

Addendum: split GPU-v3 failed-host attempt

Separate attempt, 05:41:00–05:41:52 UTC. Source delivery `3e36dca3b10a69f013e97c965f4cec7ed1c1f523`; no original evidence or run1 finding is rewritten. Archive 27,213 bytes, SHA256 prefix `15516ac73ae3b4e7`. All 120 manifest-listed payload files match in retained and fresh extraction trees; 121 archived files include the manifest. Full hash and independent checks: `diagnostics/gpu-v3-audit/receipt.json`.



Build passes, device fails. CUDA13.3.73 build and host bundle exited 0: 2,862 sigma checks including48/48 known answers, 28,642 CLMUL and 28,642 portable controls. The remote native v2 positive fixture passed 8 records and 8 golden replays; it contains no zero-length report. The CUDA device doctor then exited 3: unknown error, driver API 13000/runtime 13030, devices empty. Native GPU KAT/checker, GPU walk and GPU-report ingestion did not run. Health/restart, benchmarks and challenge likewise did not run.

The host's `nvidia-smi` records RTX5090, driver 580.173.02 and **GPU Recovery Action: Reboot**. These observations identify this failed device-initialization attempt and recovery flag; they do not establish a general toolchain incompatibility or change run1's earlier successful control. All four negation commands returned 0; the three quotient arms remain censored and baseline is a collision flag, not an independently recovered scalar.

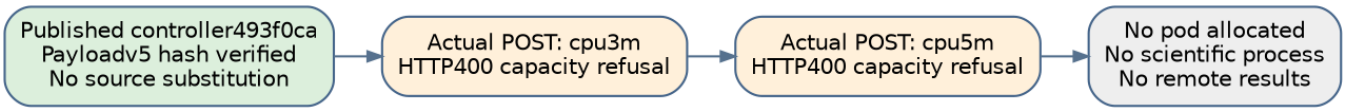
Memory gate prevents a new large closure. Physical RAM was 405,163,536,384bytes, but cgroup/effective RAM was 124,999,999,488bytes, below the revised 192,000,000,000-byte gate. Gate exit 78 records the original large n19/m3/ell7 cell as unexecuted; tiny SAT/UNSAT controls passed before that gate. No new frontier measurement or D is supplied. Run1's earlier allocation failure remains the only large-frontier attempt in this packet.

CPU-v3 capacity refused. The separate 256 GB CPU request received an actual provider HTTP500 capacity refusal, pod id null. No CPU allocation or scientific trial occurred. This is an infrastructure refusal, not evidence about the mathematics. The GPU controller fetched evidence, deleted its pod and verified absence; an independent watchdog also found absence. Reporting performed no provider or Git mutation.

Current record. See `GPU_V3_RESULTS.md` and `reports/continuation-summary.json`. The pre-addendum three-page PDF is preserved as `reports/run1-report.pdf`. Original ingestion failure, separate local v2 qualification and unreplayed-trail limits remain unchanged. Historical campaign rate figures and forecasts remain unchanged; no throughput, challenge completion, solving degree, attack gain or scientific transition is inferred from this failed-host attempt.

Addendum: CPU API-v2 capacity refusals

Two actual requests, no allocation or scientific trial. Fresh `resource-runs/cpu-api-v2-01`, observed approximately 06:03:31–06:03:35 UTC, is bound to published controller source `493f0ca1b7183dbd9e42ddcf8c44c650754e3fdb`. Both exact CPU-create requests returned HTTP400 with the explicit provider message: "There are no longer any instances available with the requested specifications."



Request	Catalog-derived RAM	Catalog availability	Actual response
cpu3m,32 vCPU	256 GB	HIGH	HTTP400 capacity refusal
cpu5m,32 vCPU	256 GB	LOW	HTTP400 capacity refusal

Catalog quotes were $1.76/hour and $2.08/hour. These are request/catalog metadata, not measurements of a 256 GB instance or of research performance. The controller stopped after the two explicit refusals, without allocating another pod, starting science or fetching remote results. Refusal does not establish general lack of capacity, mathematical failure or security.

Source and raw evidence. Redacted requests, both refusal records, catalog, ownership intent, controller stdout/stderr and original outcome are retained under the fresh resource directory and adjacent controller logs. Public-source proof records an unauthenticated fetch of 1,626,771 bytes and the unchanged payloadv5 hash prefix `e269ad82ff1ff772`; the full digest and source URL are retained in `public-source-proof.json` under that directory. This binds public delivery provenance, not a scientific execution. The outcome retains pod id null and results fetched false.

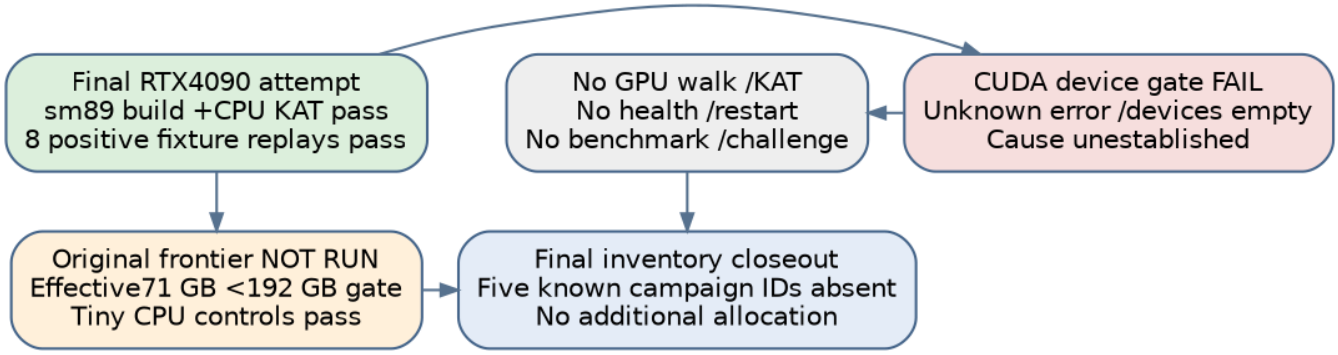
Independent no-allocation closeout. Root's separate receipt records a complete API-v2 inventory (one page), zero exact-request matches and zero active campaign resources. No provider DELETE was required. Root verified the exact local watchdog process identity/owner, stopped only that process with SIGTERM and recorded it stopped. The original controller outcome's false absence/deleted fields remain unchanged; the separate closeout binds the no-allocation observation. Receipt: `resource-runs/cpu-api-v2-01/root-no-allocation-closeout.json`.

The frozen intent watchdog could otherwise poll an empty inventory every two seconds until its deadline. Explicit closeout avoids those needless local polls; controller/scientific sources were not changed.

Evidence boundary. This attempt supplied no closure output, D, curve computation, throughput, relation yield or challenge progress. Run1's GPU control, original ingestion refusal, separate local correction and censored closure remain unchanged; GPU-v3 is a separate failed-host record. Historical campaign rates, forecasts and scientific status remain unchanged. The previous four-page report is preserved as `reports/gpu-v3-report.pdf`, SHA256 prefix `fb389bfbcf60bc1f`. Full source-linked record: `CPU_API_V2_RESULTS.md`.

Final allocated GPU attempt and campaign resource closeout

RTX4090 attempt, 06:22:18–06:24:10 UTC. Separate scope `resource-runs/gpu-final-01`, source `390898259771f708e431a49f356dcf2a296f82aa`. Catalog quote $0.34/hour, provider 36vCPU and effective 71GB RAM. The nested raw basename `remote/run-gpu-v3` is distinct through its full resource-run path.



Observed passes and failed device gate. CUDA13.3.73 compiles the client for compute89/sm89. CPU controls pass 48/48 known answers, 2,862 sigma, 28,645 CLMUL and 28,642 portable checks. The remote corrected-ingester positive fixture passes 8 records/8 golden replays; no zero-length case is in that fixture. Device initialization exits 3 with CUDA unknown error, driver API 13000, runtime 13030 and an empty device list. Native GPU KAT/checker, GPU walk, GPU-report ingestion, health/restart, throughput samples and challenge collection do not run.

`nvidia-smi` reports driver 580.126.09 and GPU Recovery Action Reboot. The cause of initialization failure remains **unknown**. That field alone is not a physical-hardware diagnosis; general toolchain incompatibility is not established. Build/CPU fixture passes are not GPU arithmetic validation.

Frontier and negation. Physical RAM 540,788,072,448bytes, effective cgroup 70,999,998,464bytes; the 192,000,000,000-byte gate exits 78. Tiny SAT/UNSAT controls pass, but the original large n19/m3/ell7,draw0,degree5 closure does not execute. No new D or memory-frontier measurement is supplied. Baseline negation reports a collision flag at 42,678 steps; the three quotient arms are censored at 2,208,379 steps. No recovered-scalar certificate or gain follows.

Evidence and cleanup. Archive 27,318 bytes, SHA256 prefix `d74c3fab830939e3`. All 120 manifest-listed payload files verify in retained and fresh trees with exact coverage; 121 archived files include the manifest. Raw records remain unchanged. Controller outcome records fetch, exact-ID deletion and absence. Root's separate complete one-page API-v2 inventory finds all five known allocated campaign IDs absent and zero campaign-prefix resources; a separate exact latest-ID GET returns 404. The detached local watchdog is state Z (stopped, not running), not claimed absent. No further allocation follows. Receipt: `resource-runs/gpu-final-01/root-final-inventory.json`.

Unchanged scope. The prior five-page PDF is preserved as `reports/CPU-api-v2-report.pdf`, hash prefix `70b17e44201e6dff`. Full record: `GPU_FINAL_RESULTS.md`. Run1's short control and local ingestion qualification remain scoped; production rates/forecasts, IC leaderboard and scientific status stay unchanged. This closes the bounded resource attempt, not the missing healthy-device, restart, benchmark, challenge-scalar or original large-closure obligations.